Jahangir Mammadli

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Education

University of Wisconsin–Madison, Madison, WI — Expected May 2028
B.S. Computer Sciences, Data Science Certificate — GPA: 3.79 | Dean's List
Transferred from Arizona State University
Coursework: Data Structures & Algorithms, Probability & Statistics, Discrete Mathematics, Exploring Data in R & Python, Computer Organization, Programming III (in progress), Linear Algebra & Optimization (in progress)

Projects

Edwin – Course Learning Platform with Bayesian Mastery Tracking — edwinlearning.com
Python, Flask, PostgreSQL, Redis, Docker, Gemini, RAG

- Built a learning platform for university courses: an AI assistant that answers from the course's own materials with citations, quizzes, a discussion board, and per-student, per-topic mastery tracking for instructors. Piloted in ASU's CSE 310 with 42 enrolled students.
- Modeled mastery as Bayesian Beta posteriors over an append-only evidence log with source weighting and recency decay, recomputed idempotently under retries; adaptive assessment picks the next topic by expected variance reduction.
- Designed a hierarchical Thompson sampling policy over 7 teaching strategies, shrinking per-student estimates toward a population prior for cold start; validated convergence on synthetic data.
- Built the retrieval layer: concept resolution, lexical reranking, citation checks against retrieved sources, and an abstention gate that labels answers outside the course materials as general knowledge.
- Migrated the backend from Snowflake to PostgreSQL (23 tables) and inference to hosted LLM APIs to cut multi-second latency; built deletion, retention and access controls for FERPA-protected data ahead of UW–Madison's security review.
- **Placed 2nd of 130+ teams (1,100+ participants) at SunHacks 2025.**

kt-bench – Benchmarking Knowledge-Tracing Models — github.com/Janik10/kt-bench
PyTorch, pandas, PyArrow, scikit-learn, pytest, GitHub Actions

- Implemented DKT (LSTM) and SAKT (self-attention) from scratch in PyTorch and benchmarked them against BKT, PFA and a Bayesian baseline on ASSISTments 2009 and an EdNet sample (47K students, 6.1M answers), using student-level held-out splits and equal tuning budgets.
- DKT reached 0.758 ± 0.005 AUC across 5 held-out test draws on ASSISTments 2009, within 0.004 of published results, and beat BKT by about 0.03 AUC on the same split.
- Built a Parquet data pipeline and CI-tested guards against label leakage and use of future answers (causality, determinism and padding tests); one command reproduces every result.

Experience

Backend Developer Intern – Yolgen (AI-driven career insights) — Remote | Jun 2025 – Aug 2025

- Developed backend APIs and database models powering AI-driven professional simulation features.
- Integrated OpenAI GPT APIs to generate real-time career feedback and adaptive recommendations.
- Implemented authentication, session management, and data pipelines for personalized insights.
- Reduced median API response time ∼30% through query optimization and caching, working in a 5-person agile team.

Technical Skills

Machine Learning: PyTorch, Bayesian inference, Thompson sampling / bandits, knowledge tracing, model evaluation and calibration, RAG
Languages: Python, SQL, TypeScript, JavaScript, Java, C
Backend & Data: Flask, FastAPI, Node.js, PostgreSQL, Redis, Pandas, NumPy, Parquet
Infrastructure: Docker, AWS, Linux, Git, GitHub Actions, React, REST APIs